Set Notation & Venn Diagrams

Exam code: 0607

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Written by: Mark Curtis
Reviewed by: Dan Finlay

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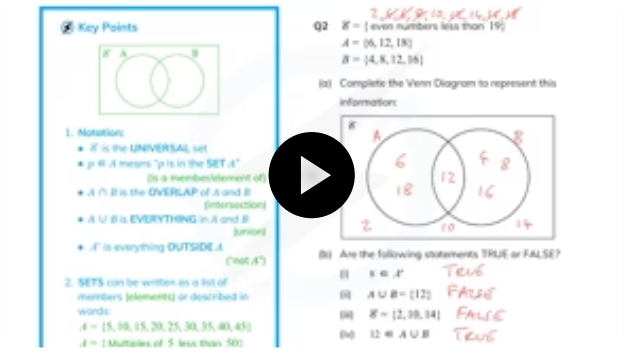

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Set notation

What is a set?

- A set is a **collection of elements**
  - Elements could be anything
    - Numbers, letters, coordinates, ...
- You could describe a set by writing its elements inside **curly brackets** {}
  - {1, 2, 3, 6} is the set of factors of 6
- If the set of elements **follow a rule** then you can write this using a **colon inside the curly brackets** {... : ...}
  - The bit before the colon is the type of element
  - The bit after the colon is the rule
    - {x is a positive integer : $x^2 < 30$} is the set of positive integers which, when squared, are less than 30
    - This is equal to {1, 2, 3, 4, 5}
  - The colon is often read as '**such that**'
  - If **no type** is specified, x can take **any value** (fractions, decimals, irrationals, ...)
    - {x: $x^2 < 30$} means any value whose square is less than 30
  - A colon can also be replaced by a **vertical bar**
    - {$x \mid x^2 < 30$}

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- $\mathscr{E}$ is the **universal** set (the set of **everything**)
  - For example, if we are only interested in factors of 24 then $\mathscr{E}$ = {1, 2, 3, 4, 6, 8, 12, 24}
  - You may see alternative notations used for $\mathscr{E}$
    - U is a common alternative (different to $\cup$ for union!)
    - S or the Greek letter ξ (xi) may also be seen

- We use **upper case** letters to represent **sets** (A, B, C, ...) and **lower case** letters to represent **elements** (a, b, c, ...)
- n(A) is the **number of elements** in set A
  - For example, if $\mathscr{E}$ = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {1, 4, 9}, B = {1, 2, 3, 4, 5, 6}
    n(A) = 3, n(B) = 6

- $A \cap B$ means the **intersection** of A and B (the **overlap** of A **and** B)
  - This is the set of elements that are in **both** set A and set B
    $A \cap B$ = {1, 4}

- $A \cup B$ means the **union** of A and B (**everything** in A **or** B **or** both)
  - This is the set of elements that are in **at least one** of the sets
  - This **includes** elements in both sets (in the intersection)
    $A \cup B$ = {1, 2, 3, 4, 5, 6, 9}

- A' means the **complement of *A***
  - It is the **set** of all elements in the universal set $\mathscr{E}$ that are **not in *A***
    A' = {2, 3, 5, 6, 7, 8, 10}

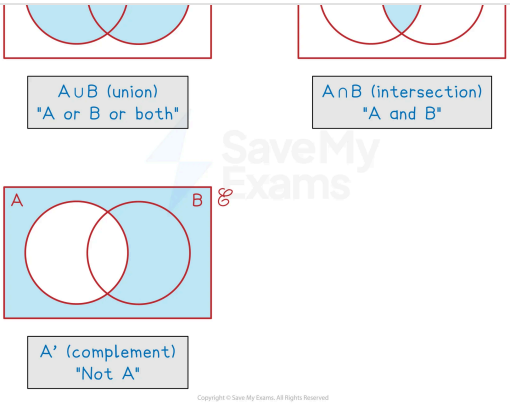
Sets & venn diagrams

What is a Venn diagram?

- A Venn diagram is a way to illustrate **all the elements within sets** and any **intersections**
- A Venn diagram consists of
  - a **rectangle** representing the **universal set ($\mathscr{E}$)**
  - a **circle** for each **set**
    - Circles may or may not overlap depending on which **elements** are shared between **sets**

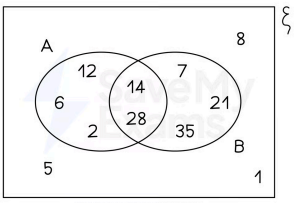
What do the different regions mean on a Venn diagram?

- $A \cup B$ is represented by the regions that **are in** A or B or both
- $A \cap B$ is represented by the region where the A and B circles **overlap**



Worked Example

Two sets A and B are shown in the Venn diagram.



(a) Write down n(A).

Answer:

The elements of A are anything inside the A circle

$$A = \{2, 6, 12, 14, 28\}$$

n(A) means the number of elements in A
There are 5 elements in A

$$\mathbf{n(A) = 5}$$

(b) Use set notation to complete the sentence {14, 28} = ...

Answer:

14 and 28 are the elements that are both in A and B
This means they are in the intersection of A and B

$$\mathbf{\{14, 28\} = A \cap B}$$

(c) Write down the elements that are in set $A \cup B$.

Answer:

$A = \{2, 6, 12, 14, 28\}$ and $B = \{7, 14, 21, 28, 35\}$
$A \cup B$ is the set of elements that are in at least one of the sets
For elements in both, only write them out once

$$\mathbf{A \cup B = \{2, 6, 7, 12, 14, 21, 28, 35\}}$$

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